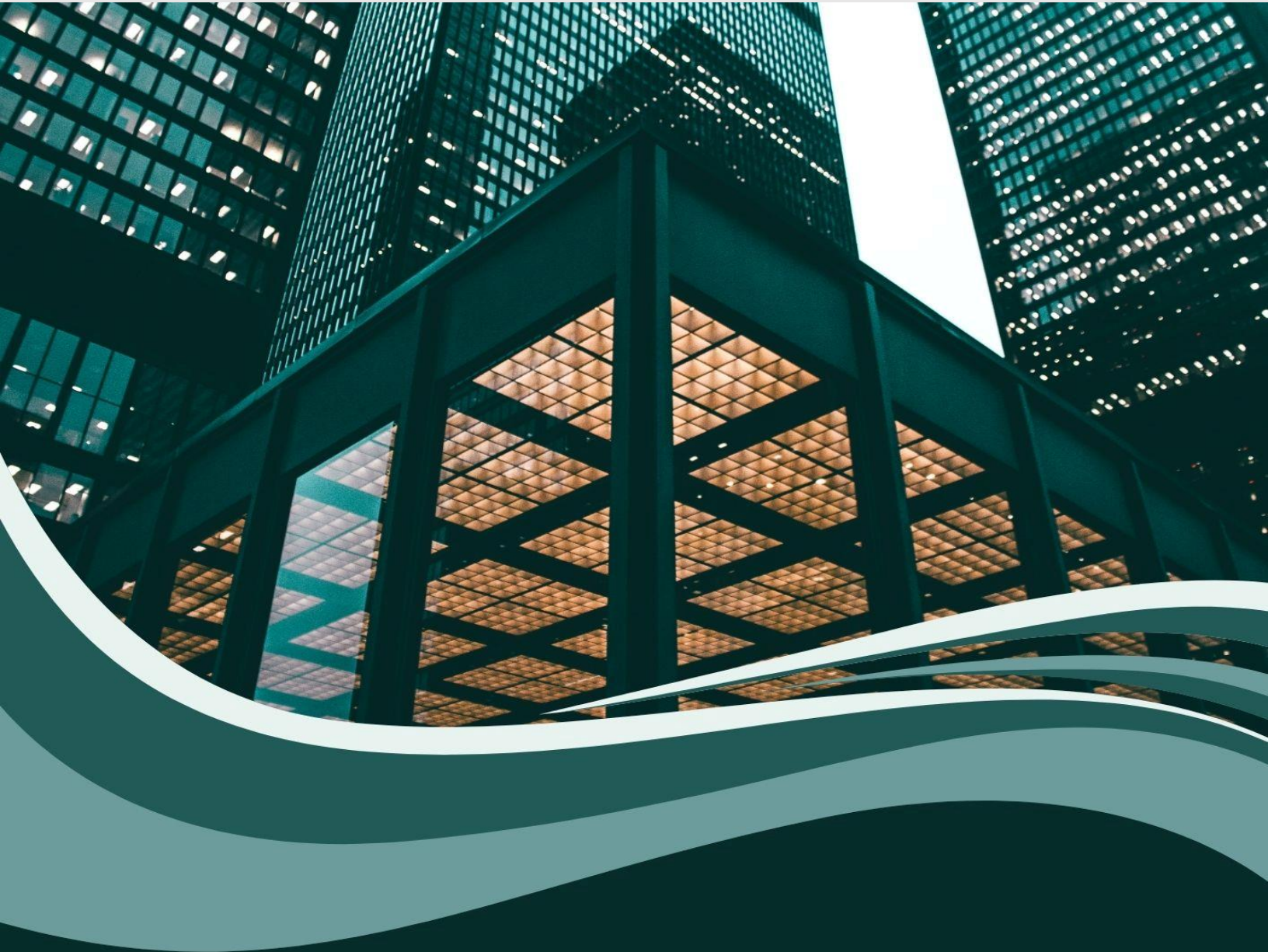




• 2025 •

REPORT

Presented to:
Digital Egypt Pioneers Initiative - DEPI

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Report: Brazilian E-Commerce Data Analysis

1. Data Cleaning and Preprocessing

1.0 Data Preparation Using Python (Google Colab)

- **Environment Used:** Google Colab with Pandas, NumPy, and Matplotlib.
- **Data Import:** Loaded multiple datasets (customers, orders, products, sellers, payments, etc.).
- **Data Cleaning Steps:**
 - Handled missing values using imputation or removal where necessary.
 - Standardized column formats (date-time conversion, numerical adjustments).
 - Detected and removed duplicate records.
- **Feature Engineering:**
 - Created new calculated fields: Total Revenue, Net Profit, Gross Margin%.
 - Extracted Month-Year from date fields for trend analysis.
- **Data Export:** Cleaned data was saved in CSV format and imported into Power BI for further analysis.

1.1 Data Acquisition and Integration

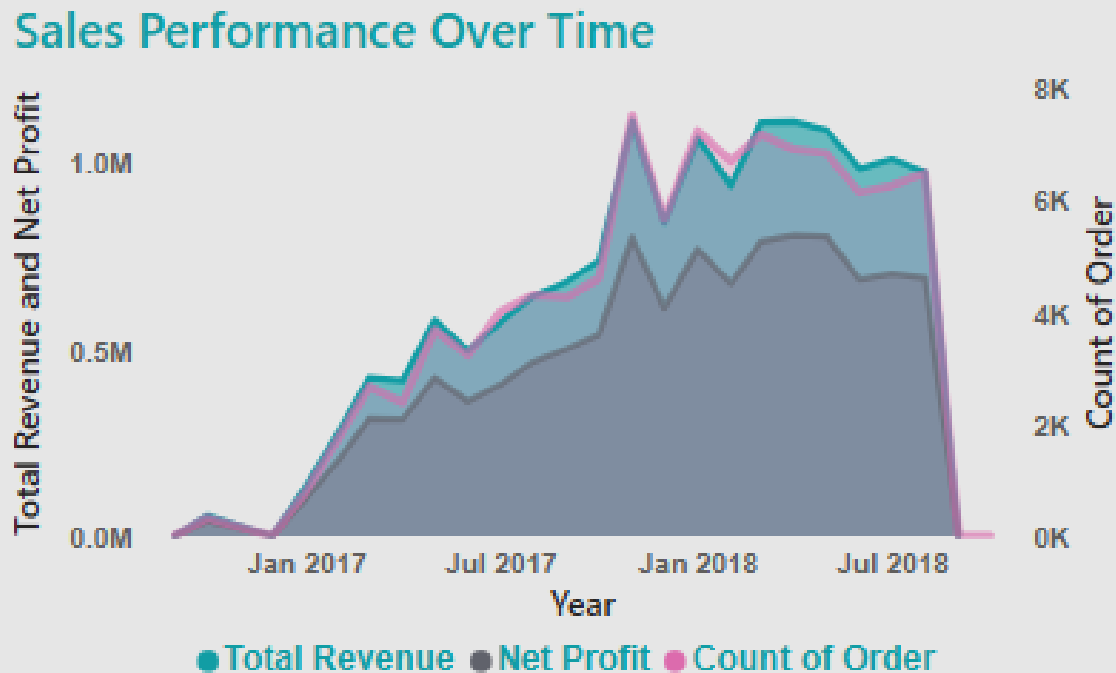
- **Datasets Loaded:** Customers, Geolocation, Order Items, Order Payments, Order Reviews, Orders, Products, and Sellers.
- **Data Merging:** Created a unified dataset by merging relevant tables on `order_id` to ensure a holistic analysis.
- **Column Selection:** Retained key attributes such as Order Date, Payment Details, Order Status, Product Prices, and Review Scores.

1.2 Data Transformation and Quality Checks

- **Missing Value Treatment:** Applied appropriate imputation techniques and handled null values.
- **Data Type Standardization:** Converted date fields to `datetime` format for temporal analysis.
- **Derived Metrics:** Introduced calculated fields, including:
 - **Total Revenue:** `price + freight_value`
 - **Net Profit:** `price - freight_value`
 - **Gross Margin %:** `(Net Profit / Total Revenue) * 100`

2. Key Analytical Findings and Business Insights

2.1 Sales Performance Evaluation (From Overview Dashboard)

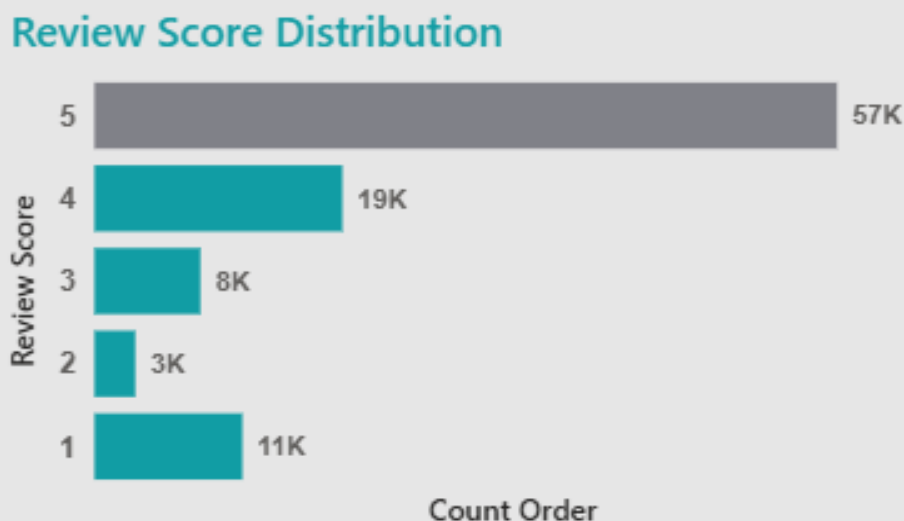


- **Total Revenue: \$15.84M** from successfully delivered orders.
- **Total Orders: 99.44K** orders processed.
- **Average Order Value (AOV): \$144.07**, indicating stable customer spending behavior.
- **Net Profit: \$11.06M**, achieving a **58.13% Gross Margin**.

2.2 Payment Accuracy and Financial Reconciliation

- Identified potential revenue variances between recorded revenue and payment collections, requiring reconciliation improvements.

2.3 Order Breakdown and Trends (From Overview Dashboard & Monthly Performance)



- **Order Volume Trends:** Orders peaked during **mid-2017 and early 2018**.
- **Revenue Forecast:** Projected revenue fluctuations with expected monthly variations.
- **Review Score Distribution:**
 - **5-star ratings:** 57K orders.
 - **4-star ratings:** 19K orders.
 - **Lower ratings (1–3 stars):** Indicate potential customer dissatisfaction points.

2.4 Delivery Performance Metrics (From Overview Dashboard)

- Identified **regional disparities in delivery performance**, with delayed shipments impacting customer satisfaction.

2.5 Profitability Analysis and Revenue Forecasting (From Monthly Performance Page)

- **Average Net Profit per Order:** \$2,311.36.
- **Total Net Profit:** \$11.85M.
- **Gross Margin Average:** 58.13%.
- **Seasonal Trends:**
 - Revenue peaks in **December 2017 and September 2018**.
 - Monthly profitability trends highlight variations due to promotions and seasonal demand shifts.

3. Power BI Dashboard Insights

The Power BI dashboard provides **three key pages**:

3.1 Overview Dashboard (High Level Business Summary)

- **Sales Performance Over Time:** Total revenue trends and order count analysis.
- **Revenue Forecast:** Predicts revenue changes using historical trends.
- **Review Score Distribution:** Breakdown of customer feedback across different rating levels.

3.2 Product Sales Insights (Granular Analysis of Product Categories)

Product Category Name	Price	Freight Value	Total Revenue	Count of order
beleza_saude	1,258,681.34	182,566.73	1,441,248.07	8836
relogios_presentes	1,205,005.68	100,535.93	1,305,541.61	5624
cama_mesa_banho	1,036,988.68	204,693.04	1,241,681.72	9417
esporte_lazer	988,048.97	168,607.51	1,156,656.48	7720
informatica_acessorios	911,954.32	147,318.08	1,059,272.40	6689
moveis_decoracao	729,762.49	172,749.30	902,511.79	6449
cool_stuff	635,290.85	84,039.10	719,329.95	3632
utilidades_domesticas	632,248.66	146,149.11	778,397.77	5884
automotivo	592,720.11	92,664.21	685,384.32	3897
ferramentas_jardim	485,256.46	98,962.75	584,219.21	3518
brinquedos	483,946.60	77,425.95	561,372.55	3886
bebes	409,830.89	68,326.11	478,157.00	2884
perfumaria	399,124.87	54,213.84	453,338.71	3162
telefonias	323,667.53	71,215.79	394,883.32	4199
moveis_escritorio	273,960.70	68,571.95	342,532.65	1273
papelaria	230,943.23	46,798.48	277,741.71	2311
pcs	222,963.13	9,836.30	232,799.43	181
pet_shop	214,315.41	39,561.24	253,876.65	1710
instrumentos_musicais	191,498.88	18,638.49	210,137.37	628
eletroportateis	190,648.58	16,020.25	206,668.83	630
eletronicos	160,246.74	46,578.32	206,825.06	2550
consoles_games	157,465.22	19,828.02	177,293.24	1062
fashion_bolsas_e_acessorios	152,823.54	31,450.00	184,273.54	1864
construcao_ferramentas_construcao	144,677.59	20,650.41	165,328.00	748
malas_acessorios	140,429.98	30,445.23	170,875.21	1034
eletrodomesticos_2	113,317.74	10,600.18	123,917.92	234
Total	13,410,174.42	2,223,712.73	15,633,887.15	97276

- **Top Revenue Generating Categories:**
 - **Beauty & Health:** \$1.44M revenue, 8,836 orders.
 - **Religious Gifts:** \$1.3M revenue, 5,624 orders.
 - **Bed, Bath & Table:** \$1.24M revenue, 9,417 orders.
- **Freight Costs Impact:** Certain categories, such as **electronics and home appliances**, have higher freight costs, impacting net margins.

3.3 Monthly Performance (Tracking Profitability and Growth)

- **Net Profit and Gross Margin Trends:** Monthly profitability tracking to evaluate performance fluctuations.
- **Seasonal Sales Analysis:** Key revenue months and fluctuations due to external factors.
- **Cancellation Trends:** Identifies areas where cancellations impact revenue flow.

4. Strategic Recommendations

4.1 Operational Enhancements

- Implement predictive analytics to reduce late deliveries and improve logistics efficiency.
- Address cancellation reasons through proactive customer engagement strategies.

4.2 Financial Optimization

- Strengthen reconciliation mechanisms to reduce revenue leakages.
- Improve payment collection strategies to enhance cash flow management.

4.3 Customer Centric Strategies

- Utilize customer review analysis to identify service gaps and improve user experience.
- Deploy targeted marketing efforts based on purchasing patterns and seasonal demand.

5. Next Steps and Implementation Plan

- Refine Power BI Dashboards for real time business monitoring and trend tracking.
- Conduct in depth customer segmentation for better targeted marketing strategies.
- Optimize supply chain operations to minimize order cancellations and delivery delays.